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ROUTING AT NETWORK LAYER

a) Simulate Static Routing Protocol Configuration using Cisco Packet Tracer.

Aim: To simulate static Routing Protocol Configuration using Cisco Packet Tracer.

PROCEDURE:

1. Open Cisco Packet Tracer
2. Place two routers two switches and two PCs
3. Connect static routes on devices using suitable cables
4. Assign IP addresses to the PCs and router interface
5. Configure static route on both routers.
6. Enable the router interface using no shutdown
7. Test connectivity between the PCs using the ping command.

STATIC ROUTING:

Static Routing is a networking method where a network administrator manually creates and configures fixed paths in a router's routing table

IP configuration

Device	IP address	Subnet Mask
PC0	192.168.1.2	255.255.255.0
Router 1 LAN	192.168.1.1	255.255255.0
Router 2 LAN	192.168.2.1	255.255.255.0
Router 1 Link	10.0.0.1	255.255.255.252
Router 2 Link	10.0.0.2	255.255.255.252
PC1	192.168.2.2	255.255.255.0

CONFIGURATION COMMANDS

Static Routing

Router 1

```
enable
configure terminal
ip route 192.168.2.0 255.255.255.0
10.0.0.2
end
```

Router 2

```
enable
configure terminal
ip route 192.168.1.0 255.255.255.0
10.0.0.1
end
```

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8. RIP ROUTING

Aim:

To simulate RIP (Routing Information Protocol) using Cisco Packet Tracer.

PROCEDURE:

1. Open Cisco Packet Tracer and create the network topology using two routers, two switches and two PCs
2. Connect the devices using suitable cables.
3. Assign IP addresses to the PCs and the router interfaces.
4. Enable the router interfaces using no shutdown
5. Configure RIP version 2 on both routers.
6. Advertise the directly connected networks using the network command.
7. Test connectivity between the PCs using ping

RIP ROUTING:

RIP (Routing Information Protocol) is a legacy, interior gateway routing protocol used to automatically manage and discover paths within a small local network.

INPUT:

INPUT (PC0 Command Prompt)

ping 192.168.2.2

OUTPUT:

Reply from 192.168.2.2 : bytes = 32
Reply from 192.168.2.2 : bytes = 32
Reply from 192.168.2.2: bytes = 32
Reply from 192.168.22 : bytes : 32

RESULT:

Thus static routing was configured and tested successfully using Cisco Packet Tracer.

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13. RIP ROUTING

Aim:
To simulate RIP (Routing Information Protocol) using Cisco Packet Tracer.

PROCEDURE:

1. Open Cisco Packet Tracer and Create the network topology using two routers, two switches and two PCs
2. Connect the devices using suitable Cables
3. Assign IP addresses to the PCs and the router interfaces.
4. Enable the router interfaces using no shutdown
5. Configure RIP version 2 on both routers.
6. Advertise the directly connected networks using the network Command.
7. Test Connectivity between the PCs using ping

RIP ROUTING:

RIP (Routing Information Protocol) is a legacy, interior gateway routing protocol used to automatically manage and discover paths within a small Local network.

CONFIGURATION

Router 1

```
enable
configuration terminal
router rip
 version 2
 no auto-summary
 network 192.168.1.0
 network 10.0.0.0
 end
```

Router 2

```
enable
Configure terminal
router rip
version 2
no auto-summary
network 192.168.2.0
network 10.0.0.0
end
```

INPUT:

INPUT (PC0 Command Prompt)

ping 192.168.2.2

OUTPUT:

Reply from 192.168.2.2: bytes = 32
Reply from 192.168.2.2: bytes = 32
Reply from 192.168.2.2 bytes = 32
Reply from 192.168.2.2 bytes = 32

RESULT:

Thus RIP routing was configured and tested Successfully using Cisco Packet Tracer.